Choline-Geranate Ionic Liquids for Adipose Tissue Ablation and Minimally Invasive Treatment of Lipomatous Tumors

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Liposarcoma, one of the most common soft tissue sarcomas, can be extremely challenging to treat. While surgery and chemotherapy have been successful in less severe cases, effective treatment is still limited due to post-resection recurrence rate, treatment toxicity, and the complexity of the tumor. Here, a biocompatible ionic liquid consisting of choline and geranic acid (CAGE IL) is characterized and tested for diffusion, bactericidal, and ablative properties. In an adipose mimicking gel matrix, CAGE IL demonstrated uniform radial diffusion. 3T3-L1 mouse adipocytes and adipose tissue in subcutaneous canine skin models show significant ablation after treatment. Furthermore, *E. coli* showed substantial susceptibility to CAGE IL, suggesting that IL-based treatments would be able to prevent infectious complications. With its abilities to prevent infection, evenly diffuse through fat, and ablate adipose tissue, CAGE IL offers a promising alternative to fatty tissue tumor treatments while reducing risks of recurrence and exposure to toxicity.